Amr Walid Maher

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PROFESSIONAL SUMMARY

Data Engineer and Full-Stack Developer and Cairo University Economics graduate (2026), who ships complete, production-grade systems — not just prototypes. I design robust **ETL/ELT pipelines, data warehouses and lakehouses** (Python, SQL, Spark, Kafka, Azure) and build full **business applications** (POS, CRM, asset tracking, e-commerce) with **ASP.NET Core 8 / C#** and **Next.js / React**, delivered with Docker, CI/CD and clean layered architecture. I combine data-engineering rigor with software-engineering discipline — validation, testing, audit logging and security — to turn business problems into reliable systems that run in production.

EDUCATION

Bachelor's Degree in Economics (Graduated) 10/2022 – 06/2026

Faculty of Economics and Political Science – Cairo University

- GPA: 3.67 / 4.0

EXPERIENCE

Software Engineer | Uni Group 10/2025 – Present

- Building the internal software, data and automation behind a multi-company group — from requirements to production.
- Designed and built the multi-tenant **UniGroup CRM** on ASP.NET Core (Clean Architecture) with role-based access and per-unit data isolation.
- Delivered business systems end to end: **Retail POS** (179/179 tests passing), **Asset Tracking & Support Desk**, and an **E-Commerce / CMS** platform.
- Automated WhatsApp marketing campaigns with **RFM customer segmentation**; engineered a social-media governance system (PostgreSQL, n8n, Metabase).
- Recognised as **Best Employee of the Month** (August 2026).

Data Analyst & BI Developer (Project-based) | Kidana Development Co 03/2026 – 05/2026

- Developed a BI solution using Excel (Power Query, Power Pivot, DAX) to centralize and analyze payment and bank guarantee data for the finance department.
- Delivered two interactive dashboards that enabled the team to shift from manual reporting to self-service analytics with real-time visibility and proactive risk management.

Microsoft Data Engineer Trainee | Digital Egypt Pioneers Initiative (DEPI) · MCIT 11/2025 – 07/2026

- Completed the national Microsoft Data Engineer track; built ETL/ELT pipelines with SQL, Python and Spark on the Azure data stack (ADLS Gen2, Delta Lake).
- Capstone: **CryptoPulse** — a real-time crypto lakehouse with Kafka, Spark, dbt and Airflow. Awarded a **Certificate of Achievement**.

Data Engineering Instructor | Applied Statistics Lab (ASL) · FEPS, Cairo University 05/2026 – 08/2026

- Taught an intensive program in SQL, Python, database management and ETL/ELT architecture, guiding students from data analysis to building data systems.
- Mentored students through hands-on projects; awarded **Outstanding Instructor — Certificate of Excellence** and a Certificate of Appreciation.

Data Analyst Intern | EMAM Organization for Development and Culture 02/2025 – 10/2025

- Conducted statistical research on environmental data, applying regression and hypothesis testing to identify key trends.
- Prepared data-driven reports and visualizations to support policy recommendations and ensure data quality.

Trainee | Central Agency for Public Mobilization and Statistics (CAPMAS) 07/2025 – 08/2025

- Gained hands-on experience across the full data lifecycle at Egypt's national statistical agency, from survey design to statistical reporting.

PROJECTS

CryptoPulse — Real-Time Crypto Lakehouse on Azure

Data Engineer · Kafka, Spark, Delta Lake, Azure ADLS Gen2, Airflow, dbt, FastAPI, Next.js, Docker

- Unified 4 live and historical streams (Binance, CoinGecko, NewsAPI, RSS) through **Kafka** and **Spark Structured Streaming** into a Medallion lakehouse on Azure.
- Added FinBERT sentiment scoring, a **dbt** Gold layer and **Airflow** DAGs; served data via FastAPI and a live Next.js dashboard (~15 Docker services).

Retail POS & Reporting System

Full-Stack Developer · ASP.NET Core 8, C#, EF Core, SQL Server, Identity, xUnit

- Built an Arabic (RTL) point-of-sale with barcode/camera scanning, discount codes, returns, thermal receipts and CODE-128 label printing.
- Role-based access (Admin/Agent), full audit log, live dashboard and CSV reports; shipped with **179/179 passing tests** and a 0-error build.

Asset Tracking & Technical Support System
Full-Stack Developer · ASP.NET Core 8, EF Core, Hangfire, SignalR, Serilog, ClosedXML

- Tracks the full asset lifecycle, custody, preventive maintenance and support tickets with per-company isolation via EF Core Global Query Filters.
- Real-time SignalR notifications, Hangfire background jobs, QR asset tags and Arabic Excel report export on a clean 4-layer architecture.

End-to-End SQL Server Data Warehouse
Data Engineer · SQL Server, T-SQL, Medallion Architecture, Star Schema

- Built a complete warehouse integrating ERP and CRM data with a Medallion architecture and T-SQL ETL pipelines into a **Star Schema** data mart.

AI-Powered Enterprise Analytics Dashboard
Full-Stack / Data · Next.js, React 19, Vercel AI SDK, Google Gemini, Recharts, TypeScript

- Arabic (RTL) analytics dashboard with a streaming **Gemini AI assistant** that answers questions about the underlying sales data in real time.

FTD TechZone — E-Commerce & Corporate CMS Platform
Full-Stack Developer · ASP.NET Core MVC, C#, EF Core, SQL Server, Identity

- Bilingual store + corporate site with a full admin back-office: products, dynamic attributes, orders, installments, careers, news and events.

WhatsApp Marketing & Customer Analytics System
Automation · C# / .NET, JavaScript, SQL, RFM Segmentation

- RFM-segmented campaigns with throttled, anti-ban sending and order-funnel tracking to target high-value customers instead of blind blasting.

Real-Time Stock Data Pipeline
Data Engineer · Python, MongoDB Change Streams, FastAPI, Docker Compose

- Event-driven pipeline where MongoDB Change Streams trigger aggregation analytics exposed through FastAPI, fully containerized.

M365 Automated Project Management & Attendance System
Automation / BI · Microsoft Forms, Power Automate, SharePoint, Power Query, DAX

- Fully automated project lifecycle and attendance tracking with validation flows, email alerts and a live DAX-driven KPI dashboard.

Automated Crypto Price ETL & Dockerized PostgreSQL Pipelines
Data Engineer · Python, Pandas, GitHub Actions, Docker, PostgreSQL

- Hourly CoinGecko ETL running 24/7 on **GitHub Actions**, plus a reproducible Python → PostgreSQL ETL launched with a single **docker compose up**.

TECHNICAL SKILLS

Data Engineering:	ETL/ELT Pipelines, Data Modeling, Data Warehousing, Medallion / Lakehouse, Kafka, Spark, Airflow, dbt, Delta Lake
Languages:	Python (Pandas, PySpark), SQL / T-SQL, C#, TypeScript / JavaScript, PHP
Backend & Full-Stack:	ASP.NET Core 8, EF Core, Clean Architecture, REST APIs, FastAPI, Next.js / React, Identity & RBAC, Multi-Tenancy
Databases & Cloud:	SQL Server, PostgreSQL, MongoDB, Redis, Azure (ADLS Gen2), Google BigQuery, Microsoft Fabric
DevOps & Automation:	Docker & Docker Compose, GitHub Actions (CI/CD), Git, Power Automate / M365, n8n, Playwright
BI & Visualization:	Power BI, Excel (Advanced), Power Query, Power Pivot, DAX, Tableau, Looker Studio, Metabase

CERTIFICATIONS & AWARDS

- **Best Employee of the Month** — Uni Group (August 2026)
- **Outstanding Instructor — Certificate of Excellence** — Applied Statistics Lab, FEPS, Cairo University (July 2026)
- **Microsoft Data Engineer — Certificate of Achievement** — Digital Egypt Pioneers Initiative (DEPI), MCIT (2026)
- **Certificate of Appreciation** — Applied Statistics Lab, FEPS, Cairo University (July 2026)
- **Get Started with Lakehouses in Microsoft Fabric** — Microsoft Learn (January 2026)
- **DP-700: Fabric Data Engineer Associate** — Microsoft Certified (In Progress)